ECE 310 Fall 2026
Midterm 1 Review

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September 28th, 2026



True/False Section: FA2019, Problem 1

(a) An LSI system specified by the following difference equation: $y[n] - \frac{1}{2}y[n-1] = x[n]$ can be causal or anti-causal. **T**/**F**

(b) The input and output relationship of an arbitrary system is completely determined by the system's unit pulse response. **T**/**F**

(c) If an LSI system is BIBO unstable, its unit pulse response $h[n]$ must be unbounded. **T**/**F**

(d) Let $h[n] = h_1[n] * h_2[n]$ be the unit pulse response (UPR) of two serial subsystems with UPR $h_1[n]$ and $h_2[n]$. If h_1 or h_2 is BIBO *unstable*, h must be BIBO *unstable*. **T**/**F**

(e) An LSI system with a finite-length impulse response can be BIBO stable or unstable. **T**/**F**

(f) A time-varying system cannot be causal. **T**/**F**

(g) Let $\sum_{n=-\infty}^{\infty} x[n]\delta[2^n u[n] - 8] = 4$. Then, $x[3] = 2$. **T**/**F**

(h) Suppose that the step response $g[n]$ of an LSI system, i.e., the output of the system to the input $x[n] = u[n]$ has a z-transform with a pole at $z = 1/2$. Then, $H(z)$ has also pole at $z = 1/2$. **T**/**F**

(i) If the response $y[n]$ of an LSI system to the input $x[n] = 3^n u[n]$ is unbounded, the system must be BIBO unstable. **T**/**F**

(j) The response $y[n]$ of a BIBO *unstable* LSI system to any non-zero input $x[n]$ is always unbounded. **T**/**F**

(c) Consider $h[n] = u[n]$

(d) May have pole-zero cancellation.

(h) $G(z) = H(z)\left(\frac{1}{1-z^{-1}}\right)$

We may have pole-zero cancellation

Identifying LTIC Systems Problem: SP2023, Problem 2

	Linear	Shift-Invariant	Causal	Stable
$y[n] = x[n] * (-1)^n u[n]$	Yes	Yes	Yes	No
$y[n] = \frac{x[n]}{x[2]}$	No	No	No	No
$y[n] = \cos^2\left(\frac{\pi}{2} n\right) x[n]$	Yes	No	Yes	Yes

Convolution Problem: FA2023, Problem 3

For each of the following parts, calculate the convolution $y[n] = x[n] * h[n]$.

(a) $x[n] = \{\underset{\uparrow}{1}, 2, 3, 2, 1\},\ h[n] = \{\underset{\uparrow}{-1}, 1\}$

(b) $x[n] = \left(\frac{3}{4}\right)^n u[n],\ h[n] = 2u[n] - u[n-1] - u[n-2]$

(a) $y = Hx$

$x \in \mathbb{R}^5$

$y \in \mathbb{R}^6$

$H \in \mathbb{R}^{6 \times 5}$

$$y = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$H \quad x$

$y[n] = \{-1, -1, -1, 1, 1, 1\}$

$\uparrow$

$n = 0$

(b) $h[n] = \{\underset{\uparrow}{2}, 1, 0\}$

$= 2\delta[n] + \delta[n-1]$

$y[n] = x[n] * h[n]$

$= x[n] * (2\delta[n] + \delta[n-1])$

$= 2x[n] + x[n-1]$

$y[n] = 2\left(\frac{3}{4}\right)^n u[n] + \left(\frac{3}{4}\right)^{n-1} u[n-1]$

Computing z-transform Problem: FA2024, Problem 5

For each of the following discrete-time signals, find its z-transform including ROC.

(a) $x[n] = (n+1)u[n-1]$

(b) $x[n] = u[n-1]u[3-n]$

(c) $x[n] = 3^{-n}u[n] + 3^{n}u[-n]$



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(a) $x[n] = \underbrace{(n-1)u[n-1]}_{v[n]=g[n-1]} + 2u[n-1]$

$g[n] = nu[n]$

$\updownarrow \mathcal{Z}$

$\frac{2z^{-1}}{1-z^{-1}}, \quad |z|>1$

$X(z) = V(z) + \frac{2z^{-1}}{1-z^{-1}}$

$V(z) = z^{-1}G(z)$

$g[n] = nu[n]$

$G(z) = -z\frac{d}{dz}\left(\frac{1}{1-z^{-1}}\right)$

$= -z(-1)(1-z^{-1})^{-2}(z^{-2})$

$= \frac{z^{-1}}{(1-z^{-1})^2}, \quad |z|>1$

$X(z) = \frac{z^{-2}}{(1-z^{-1})^2} + \frac{2z^{-1}}{1-z^{-1}}, \quad |z|>1$

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(b) $x[n] = u[n-1]u[3-n]$

(c) $x[n] = 3^{-n}u[n] + 3^{n}u[-n]$

(b) $x[n] = \{0, 1, 1, 1, 0, 0 \cdots\}$
$= \delta[n-1] + \delta[n-2] + \delta[n-3]$
$X(z) = z^{-1} + z^{-2} + z^{-3}, \quad z \neq 0$

Computing z-transform Problem: FA2024, Problem 5

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(b) $x[n] = \{0, \underset{\uparrow}{1}, 1, 1, 0, 0 \cdots\}$

$= \delta[n-1] + \delta[n-2] + \delta[n-3]$

$X(z) = z^{-1} + z^{-2} + z^{-3}, \quad z \neq 0$

right-sided

(c) $x[n] = \left(\frac{1}{3}\right)^n u[n] + (3)(3)^{n-1} u[-((n+1)-1)]$ ← left-sided

$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{3z^{-1}}{1-3z^{-1}}, \quad \frac{1}{3} < |z| < 3$

$-b^n u[-n-1] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-bz^{-1}}, \; |z| < |b|$

z-transform LTI System Response Problem: SP2023, Problem 6

Suppose the input $x[n]$ to a **causal and stable LTI system** produces output $y[n]$.

$$X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})}$$

$$Y(z) = \frac{1}{(1-\frac{1}{2}z^{-1})(1-z^{-1})(1-\frac{1}{4}z^{-1})}$$

(a) Find the transfer function $H(z)$ and its ROC.

(b) Find the impulse response $h[n]$.

(a) $H(z) = \frac{Y(z)}{X(z)} = \frac{(1-2z^{-1})(1-z^{-1})}{(1-\frac{1}{2}z^{-1})(1-z^{-1})(1-\frac{1}{4}z^{-1})} = \frac{1-2z^{-1}}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})}$



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(a) Find the transfer function $H(z)$ and its ROC.

(b) Find the impulse response $h[n]$.

(b) $H(z) = \frac{1-2z^{-1}}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})} = \frac{A_1}{1-\frac{1}{2}z^{-1}} + \frac{A_2}{1-\frac{1}{4}z^{-1}} = \frac{-6}{1-\frac{1}{2}z^{-1}} + \frac{7}{1-\frac{1}{4}z^{-1}}$

$1-2z^{-1} = A_1(1-\frac{1}{4}z^{-1}) + A_2(1-\frac{1}{2}z^{-1})$

$A_1: z=\frac{1}{2} \rightarrow A_1 = -6$

$A_2: z=\frac{1}{4} \rightarrow A_2 = 7$

$\longleftrightarrow Z$

$h[n] = -6(\frac{1}{2})^n u[n] + 7(\frac{1}{4})^n u[n]$

z-transform LTI System Response Problem: SP2023, Problem 6

Suppose the input $x[n]$ to a **causal and stable LTI system** produces output $y[n]$.

$$X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})}$$

$$Y(z) = \frac{1}{(1-\frac{1}{2}z^{-1})(1-z^{-1})(1-\frac{1}{4}z^{-1})}$$

(c) Determine the difference equation of the system.

$$H(z) = \frac{1-2z^{-1}}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})} = \frac{Y(z)}{X(z)}$$

$$Y(z)(1-\tfrac{1}{2}z^{-1})(1-\tfrac{1}{4}z^{-1}) = X(z)(1-2z^{-1})$$

$$Y(z)(1-\tfrac{3}{4}z^{-1}+\tfrac{1}{8}z^{-2}) = X(z)(1-2z^{-1})$$

$$y[n] - \tfrac{3}{4}y[n-1] + \tfrac{1}{8}y[n-2] = x[n] - 2x[n-1]$$

Causal System

$$y[n] = \tfrac{3}{4}y[n-1] - \tfrac{1}{8}y[n-2] + x[n] - 2x[n-1]$$

z-transform Stability Problem: SP2023, Problem 7

The transfer function of a causal LTI system is given below:

$$\frac{1-3z^{-1}}{1-4z^{-1}} = H(z) = \frac{z-3}{z-4}, \text{ ROC: } |z| > 4$$

$h[n] = 4^n u[n] - 3(4)^{n-1} u[n-1]$

(a) Find a bounded input $x[n]$ that will produce an unbounded output $y[n]$.

$x[n] = \delta[n] \rightarrow y[n] = \delta[n] * h[n] = h[n]$ → Unbounded!

(b) Find an unbounded input $x[n]$ that will produce a bounded output $y[n]$.

$X(z) = \frac{z-4}{z-3} = \frac{1-4z^{-1}}{1-3z^{-1}} \xleftrightarrow{\mathcal{Z}} 3^n u[n] - 4(3)^{n-1} u[n-1]$

$\rightarrow Y(z) = X(z)H(z) = 1 \xleftrightarrow{\mathcal{Z}} y[n] = \delta[n]$

(c) Find a bounded input $x[n]$ that will produce a bounded output $y[n]$.

$X(z) = 1 - 4z^{-1} \xleftrightarrow{\mathcal{Z}} \delta[n] - 4\delta[n-1]$

$Y(z) = X(z)H(z) = 1 - 3z^{-1} \xleftrightarrow{\mathcal{Z}} \delta[n] - 3\delta[n-1]$